Reading a W-2: which engine to build on

Findings from 17 test cases across 4 form layouts and 6 capture conditions. Scored on the four fields the NYC benefits screener actually consumes.

> ### Recommendation: Gemini Flash-Lite
>
> It scores highest (**65**), reads **94%** of fields correctly and **100%** of the income figures the screener depends on — and across all 17 cases it never once overstated income and never invented a value. It is also the fastest engine that is also accurate, at 3.2 s per document and $0.00034.

The scoreboard

ENGINE	DOCS	SCORE	ACCURACY	CRITICAL	OVER	INVENTED	BLANK	SPEED	COST/DOC
Gemini Flash-Lite RECOMMENDED	17	**65**	94%	100%	0	0	0	3.2 s	$0.00034
qwen3-vl:2b (local)	17	**38**	85%	94%	2	0	4	73 s	free
Gemini 3.5 Flash	5	**21**	96%	100%	0	0	0	6.0 s	$0.00033
Track A — OCR + parser	17	**21**	64%	67%	0	0	30	1.7 s	free

Over counts income read *higher* than the form says. **Invented** counts values returned for boxes that are blank or absent. **Blank** counts fields the engine declined to answer — cheap, because the app simply asks the user. Score weights these by real-world cost: correct +1, blank −1, understated −2, wrong −3, invented −4, **overstated −5**.

Why overstating income is the failure that matters

This pipeline pre-fills benefit applications. Read income too *low* and an extra programme appears that the user can decline — visible, recoverable, cheap. Read it too *high* and the household is pushed over an eligibility cap, the programme silently never appears, and **nobody ever finds out** — not the user, not the caseworker, and not any metric in this report. That asymmetry is why overstatement is weighted five times a correct answer.

Only one engine commits it, and it does so at high confidence:

ENGINE	TEST CASE	FIELD	OUTCOME	ON THE FORM	READ AS	CONF.
qwen3-vl:2b (local)	`laser-4up-clean`	`box1_wages`	**wrong over**	`61450.00`	`64200.00`	0.85
qwen3-vl:2b (local)	`laser-4up-phone`	`box1_wages`	**wrong over**	`61450.00`	`64200.00`	0.85

Both of qwen3-vl:2b (local)'s overstatements are the same mistake — reading Box 3 (Social Security wages) into Box 1 — reported at confidence 0.85, with no signal that anything was wrong. On a degraded capture the same model also returned `JANE DOE` and `1234 MAIN STREET, NEW YORK, NY 10001`: not a misreading but a fabricated placeholder.

How each engine behaves, case by case

Percentage of the screener's fields read correctly. Rows marked *degraded* are photographed rather than scanned — off-axis, out of focus, noisy and JPEG-compressed.

TEST CASE	GEMINI FLASH-LITE	QWEN3-VL:2B (LOCAL)	GEMINI 3.5 FLASH	TRACK A — OCR + PARSER
`adp-blur` · degraded	80	60	100	20
`adp-clean`	100	100	100	100
`adp-glare`	100	100	100	100
`adp-phone` · degraded	80	40	80	20
`crop-clean`	100	100	100	100
`gusto-clean`	100	80	—	100
`gusto-lowlight`	100	80	—	100
`gusto-phone` · degraded	60	60	—	20
`irs-redink-blur` · degraded	100	100	—	0
`irs-redink-clean`	100	100	—	100
`irs-redink-glare`	100	100	—	100
`irs-redink-lowlight`	100	100	—	100
`irs-redink-phone` · degraded	100	100	—	20
`irs-redink-skew` · degraded	100	100	—	80
`laser-4up-clean`	100	80	—	100
`laser-4up-phone` · degraded	80	40	—	0
`laser-4up-skew` · degraded	100	100	—	20

On a clean scan, the parser is its equal

Gemini Flash-Lite and Track A — OCR + parser both read **100%** of every flatbed-quality capture, on all four layouts. If users photographed their W-2 flat and in focus, the free self-hosted option would be just as good.

On a phone photo, the gap opens

Gemini Flash-Lite holds up where the others fall away. That is the case that decides it, because a phone photo is what people actually send.

What the alternatives are good at

Track A — OCR + parser — never wrong, often silent

OCR into a hand-written parser. It scored 64% overall, but every one of its 30 shortfalls is a *refusal to answer*, not a mistake: zero overstatements, zero inventions, across all 17 cases. It is also the fastest engine here (1.7 s), completely free, and the only one where the document never leaves hardware you control. It fails when OCR cannot read the printed labels, which is exactly what happens on a blurred or angled photo.

qwen3-vl:2b (local) — free and unmetered, but the risky one

Runs entirely on your own machine with no quota at all, and reads 94% of critical fields. But it is 23× slower than Gemini Flash-Lite and it is the *only* engine that overstates income. On this hardware it cannot use the GPU: a 4 GB card cannot hold its vision encoder, whose compute buffer alone requests more memory than the card has, so it runs on CPU throughout.

What this report does not claim

> **The test images were rendered, never printed**
>
> Ground truth is authored first and the image generated from it, so the truth never passes through an extractor or a transcriber — that part is sound. But no page here was printed and photographed, so genuine paper texture and true focus falloff are absent. **Every engine will score lower on real documents than it does here.** The deterministic parser has the most to lose, because it depends on OCR reading printed labels that real capture degrades first. Printing a subset and photographing it is the next measurement worth taking.

- Gemini 3.5 Flash appears with only 5 cases; its free tier caps at 20 requests per day, so it could not complete the corpus. Its numbers are indicative only and it is not a candidate.
- Confidence figures are not comparable between engines — the parser derives it from OCR and anchor quality, the models from output shape. Each is only meaningful against itself.
- Free-tier terms differ. Google's free tier trains on submitted input; that is acceptable for synthetic fixtures and needs revisiting before real tax documents are sent.

Recommended shape

Use **Gemini Flash-Lite** as the reader, with **Track A — OCR + parser** as an independent cross-check. The two fail in unrelated ways — one is a character recogniser, the other a generative model — which is the condition that makes agreement between them meaningful rather than merely correlated. Accept a value when both agree; leave the field blank for the user when they disagree or the parser abstains. A blank field costs one point; an overstated one costs five and is never noticed.

Generated from 56 scored extraction runs held in `results/raw/`. Every figure is computed at build time by `make-report.mjs`; none is transcribed by hand. Regenerate with `node make-report.mjs`.